

AKASH TIWARI

MSc-PhD Dual Degree in Physics · IIT Kanpur

✉ akasht24@iitk.ac.in

☎ +91 6266647229

in LinkedIn

🔗 GitHub

📍 IIT Kanpur, India

EDUCATION

MSc-PhD Dual Degree in Physics — Indian Institute of Technology, Kanpur	2024 – Present SPI: 8.33 (Last Sem)
B.Sc. (Higher Mathematics) — Gov. Model Science College, Rewa (M.P.)	2020 – 2023 72.24%
Class XII (MP Board) — Saraswati Shishu Mandir, Jail Marg, Rewa	2018 86.75%
Class X (MP Board) — Saraswati Shishu Mandir, Jail Marg, Rewa	2016 81.67%

RESEARCH EXPERIENCE

MSc Research Project — Prof. Mahendra Kumar Verma, Dept. of Physics, IIT Kanpur	2025 – Present
- Computed 1D & 2D energy spectra in quantum turbulence using numerical simulation methods. - Developed GPU-accelerated fast N-body simulation leveraging NVIDIA Warp for real-time particle dynamics. - Applied high-performance computing techniques for large-scale computational physics simulations.	

PROJECTS

Railway Fault Detector Vision Transformer, PyTorch Built a fault detection system for railway tracks using Vision Transformer for image classification and defect localization.
Medical Chatbot LangChain, Flask, HTML Developed a medical Q&A chatbot using LangChain for context-aware retrieval-augmented responses with a Flask web interface.
AI Image Generator Hugging Face, Diffusion Models Created an AI-powered image generation application leveraging Hugging Face pretrained diffusion models.
Image Generation with GAN GAN, PyTorch Designed and trained Generative Adversarial Networks to synthesize realistic images from latent space representations.
ECG Sensor Electronics Lab, Op-Amps, Signal Conditioning Designed and built a hardware ECG sensor circuit in the electronics lab for capturing and displaying cardiac signals.
Fast N-Body Simulation NVIDIA Warp, GPU Computing, Python Developed a GPU-accelerated gravitational N-body simulation using NVIDIA Warp for high-performance particle dynamics.

TECHNICAL SKILLS

Languages:	Python, Warp, JavaScript, HTML/CSS
ML / DL:	PyTorch, TensorFlow, LangChain, LangGraph, NumPy, Pandas, SQL
Tools:	Git, Flask, Hugging Face, NVIDIA Warp, Linux
Domains:	Computer Vision, Deep Learning, High Performance Computing, Computational Physics

ACHIEVEMENTS

IIT JAM — Qualified, AIR 160 (Feb 2024)	JEST — Qualified, AIR 263 (Mar 2024)	CUET — Qualified, Score 218 (Mar 2024)
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CERTIFICATIONS

- Structuring Machine Learning Projects** — DeepLearning.AI (Jun 2026)
- Convolutional Neural Networks** — DeepLearning.AI (Jan 2026)
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization & Optimization** — DeepLearning.AI (Jan 2026)
- Neural Networks and Deep Learning** — DeepLearning.AI (Jan 2026)
- Introduction to Data Science in Python** — University of Michigan (Dec 2025)

RELEVANT COURSEWORK

Deep Learning for Computer Vision · High Performance Computing · Computational Physics · Statistical Mechanics · Quantum Mechanics I & II · Classical Mechanics · Classical Electrodynamics · Mathematical Methods I & II · Condensed Matter Physics · Atomic, Molecular & Optical Physics · Nuclear & Particle Physics